LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation of Fly by wire control systems using Fly by Wire demonstrator at Control Lab.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation of Fly by wire control systems using Fly by Wire demonstrator at Control Lab.
5. Specific Objectives	20.4.2.1 Trainee will be able to explain principles of Aero engine control systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Fly by Wire demonstrator; Safety: Do not touch live circuits; Instructor supervision required during operation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a fly-by-wire system?
(b) How does a fly-by-wire system differ from a conventional hydraulic system?
(c) What are the key components of a fly-by-wire architecture?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Fly by Wire Systems • System Architecture Overview • Sensor and Actuator Roles • Flight Control Computer Functions • Redundancy and Safety Features	L & Ds	PPT, IB	15 min
(b)	• Operational Principles of FBW • Signal Transmission Path • Control Law Implementation • Failure Mode Analysis • Manual Override Mechanisms	L & Ds	PPT, PDF	20 min
(c)	• Comparison with Conventional Systems • Advantages of FBW Technology • Limitations and Constraints • Future Trends in FBW • Integration with Engine Control	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the Flight Control Computer in a fly-by-wire system.
(b) Describe how redundancy is achieved in fly-by-wire architectures.
(c) What happens to the control system if a sensor fails?

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of fly-by-wire systems relate to Full Authority Digital Engine Control (FADEC) systems discussed in LP 14.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Full Authority Digital Control System.	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min